

Zhexiang (Bruce) Li

M.S. Student in Computer Science, UCLA

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EDUCATION

University of California, Los Angeles

Sep 2026 - Jun 2028 (expected)

M.S. in Computer Science

University of Southern California

May 2026

B.S. in Computer Science | GPA 3.76 / 4.00 | Viterbi Dean's List, Fall 2024-Spring 2026

SKILLS

Languages Python, C/C++, JavaScript/TypeScript, SQL

Machine learning PyTorch, Triton, RDKit, Chemprop

Development FastAPI, Next.js, Supabase, Linux

EXPERIENCE

Health Intelligence Lab (HAIL) | UCLA

Current

Research with Prof. Yuzhe Yang

Working on foundation models for type 1 diabetes (T1D).

AI Engineer Intern | PMAOS / Next Big Seek

Jul - Aug 2026

- Designed bidirectional user-video interest propagation for short-video recommendation, with multi-timescale exposure budgets and fatigue controls to balance personalization with exploration.
- Developed audio-visual affect inference for 11 languages, combining evidence from both modalities with a fixed-point decision layer that abstains when evidence conflicts.
- Evaluated a 9 mm piezoelectric millirobot design and specified a PZT-steel cantilever experiment and acceptance criteria; experimental planning only.

Full-Stack Engineer | Terlingua Threads

Jan - May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, and SEO.

Co-Founder and CTO | CrAlg Capital

2025 - Mar 2026

- Built data pipelines and automated workflows for an AI-assisted housing debt-collection platform; led technical planning.

AI Engineer Intern | Piezolab Medical

Jun - Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Wrote experiment checklists and acceptance tests for vascular-care laboratory workflows.

Application Engineer Intern | Synaptics

Jun - Aug 2023

- Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I2C sensor data as MQTT topics for internal dashboards.

PUBLICATIONS

MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification 2026

Zhexiang Li. Preprint · [arXiv:2609.30613](https://arxiv.org/abs/2609.30613)

Developed lesion-supervised token routing and evaluated retained lesion evidence on ISIC 2019 under fixed token budgets.

Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval 2026

Manuscript submitted to AAAI 2027

Encoded mass-conserving fragment pairs for spectrum reranking; evaluated ranking changes with hard negatives and token-removal experiments.

Efficient Attention via Pre-Scoring: Prioritizing Informative Keys in Transformers 2025

Zhexiang Li, Haoyu Wang, Yutong Bao, David Woodruff. Preprint · [arXiv:2505.11040](https://arxiv.org/abs/2505.11040)

Proposed key pre-scoring and implemented Triton kernels. On ChatGLM with 131k-token contexts, pre-scoring reduced perplexity from 12.0 to 9.5 at a fixed interaction budget.

Game Theory in Companies 2023

S. Li, Z. Li, Z. Liu. Advances in Economics, Management and Political Sciences, 16(1), 357-366. [DOI](#)

Modeled game-release decisions and enumerated Nash equilibria using Python and Gambit.

SELECTED PROJECTS

Fridge Manager 2026

Built a meal planner with Next.js and Supabase that generates meal and shopping plans from available ingredients, dietary preferences, and cooking time.

AI Business Card Hub 2026 - Present

Built an iOS business-card organizer with image parsing and OCR fallback, multi-photo review, and searchable contact and company records. [Code](#)